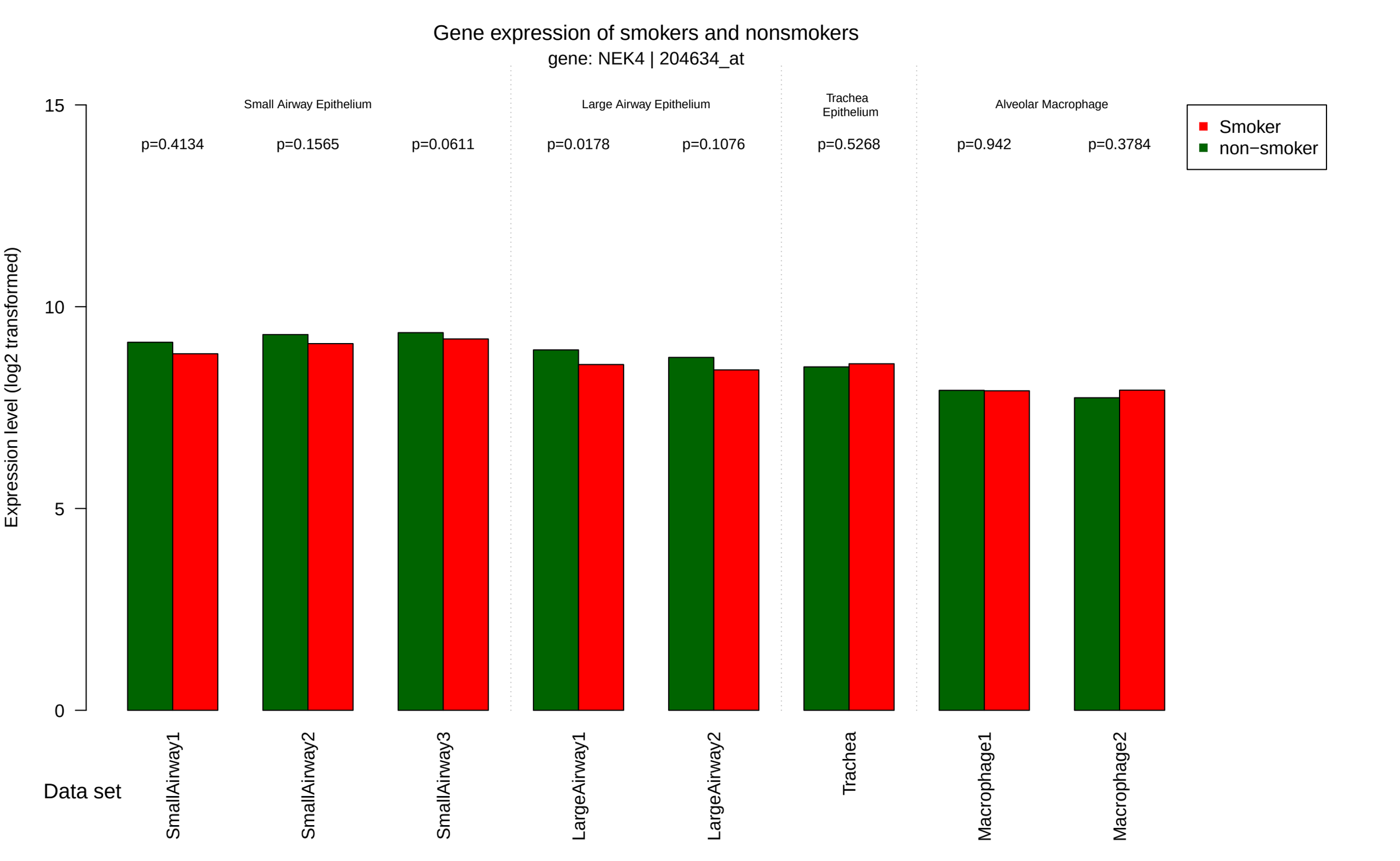
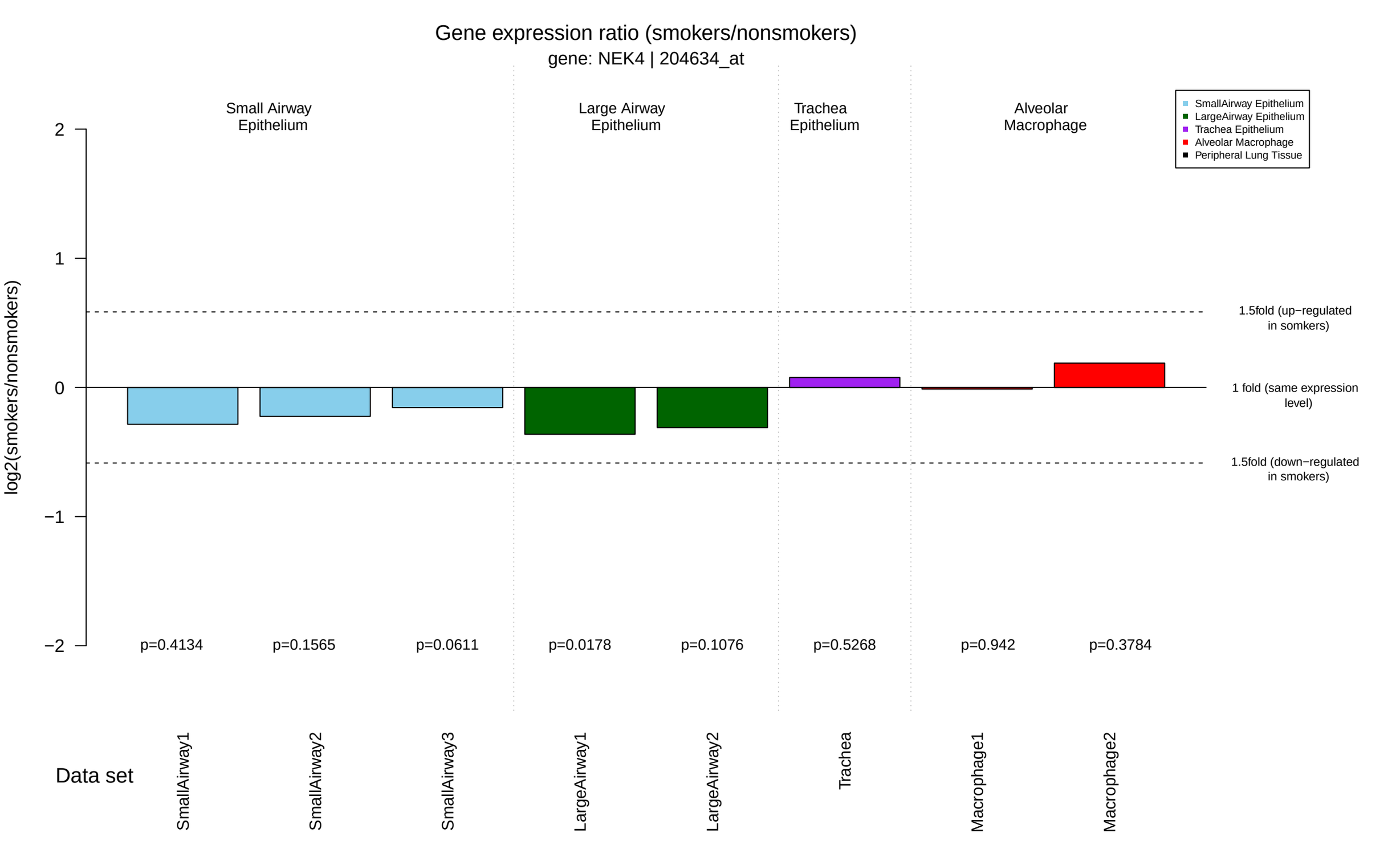
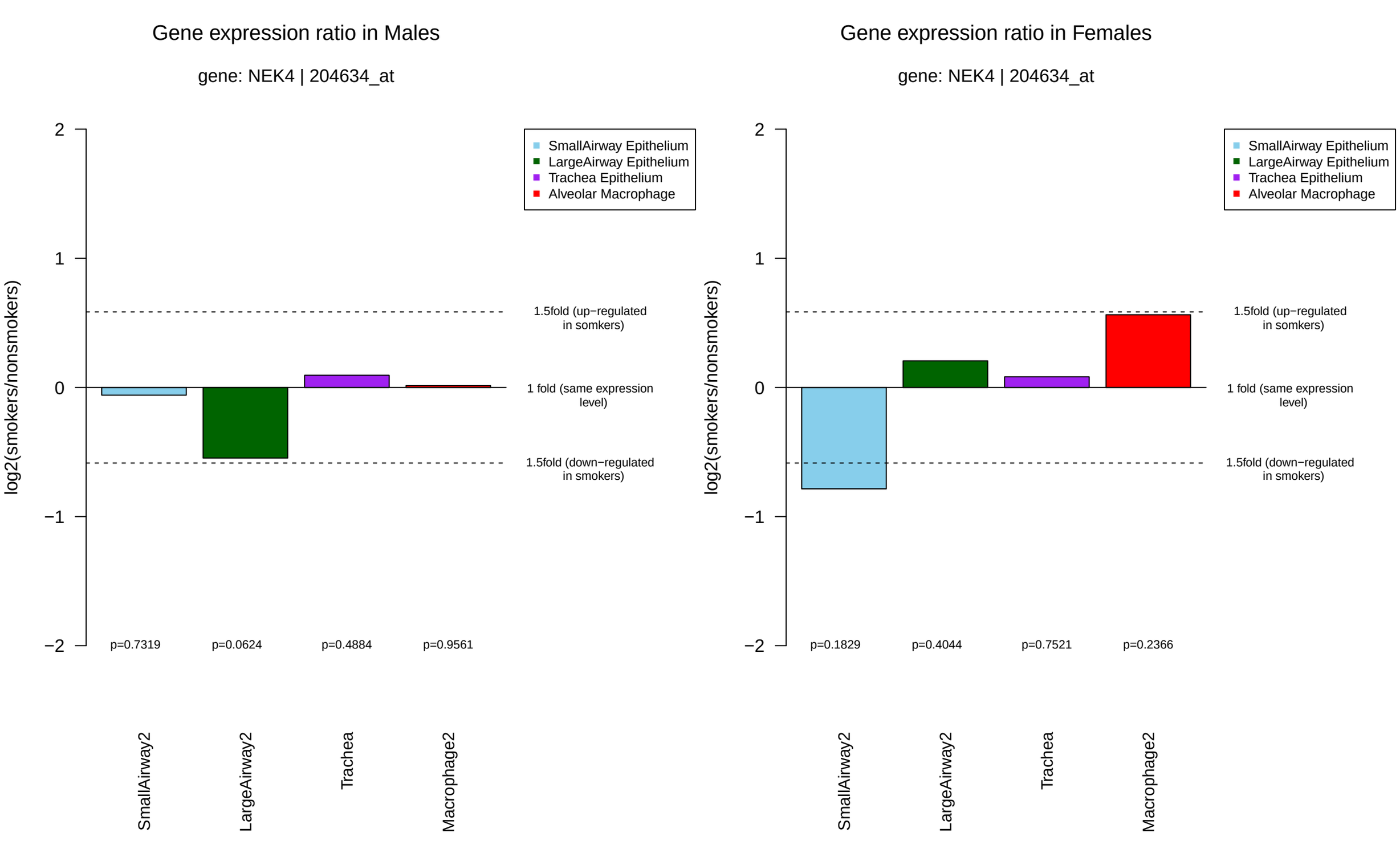
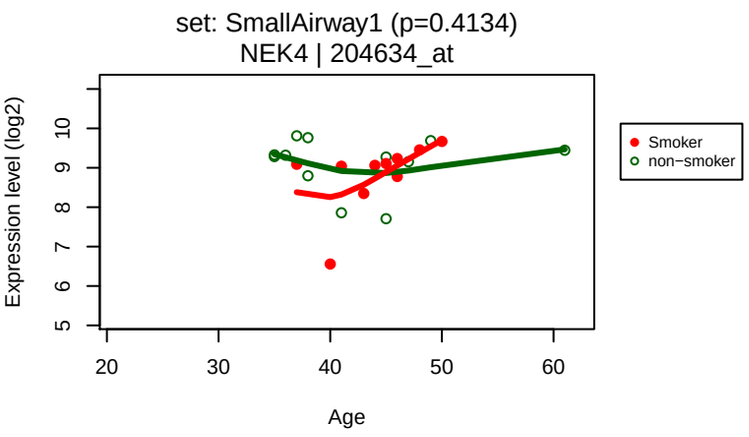
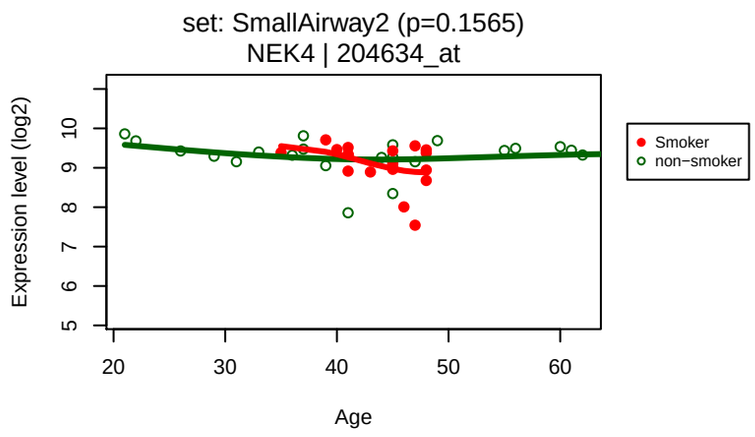
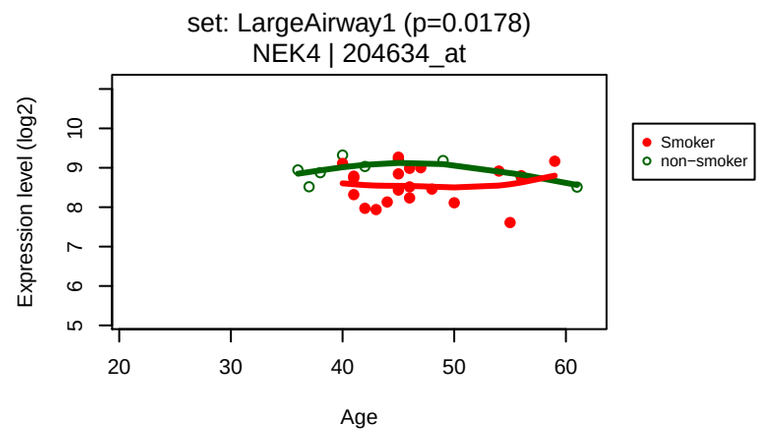

Gene expression of smokers and nonsmokers
gene: NEK4 | 204634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4134
p=0.1565
p=0.0611
p=0.0178
p=0.1076
p=0.5268
p=0.942
p=0.3784
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

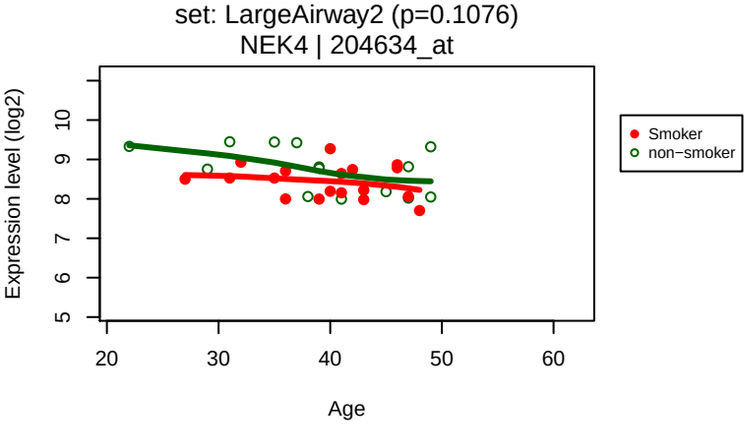

Gene expression ratio (smokers/nonsmokers)
gene: NEK4 | 204634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4134
p=0.1565
p=0.0611
p=0.0178
p=0.1076
p=0.5268
p=0.942
p=0.3784
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

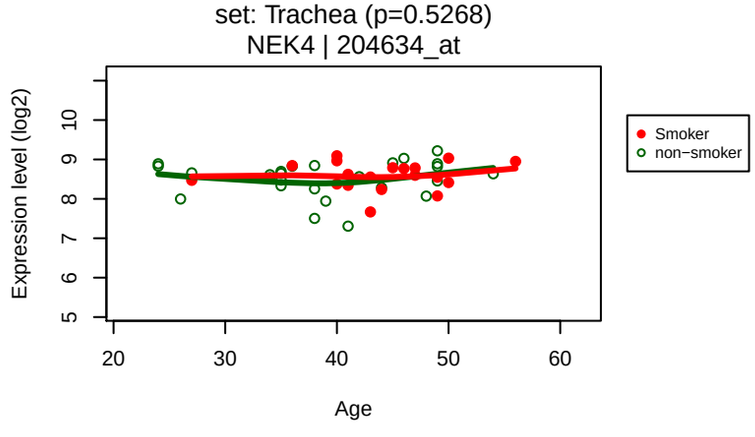

Gene expression ratio in Males
gene: NEK4 | 204634_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7319
p=0.0624
p=0.4884
p=0.9561
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK4 | 204634_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1829
p=0.4044
p=0.7521
p=0.2366
SmallAirway2
LargeAirway2
Trachea
Macrophage2

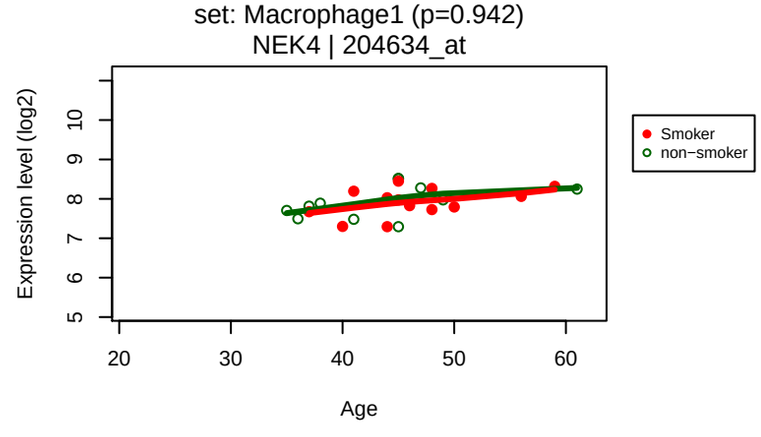

set: SmallAirway1 (p=0.4134)
NEK4 | 204634_at
Expression level (log2)
Age
Smoker
non−smoker

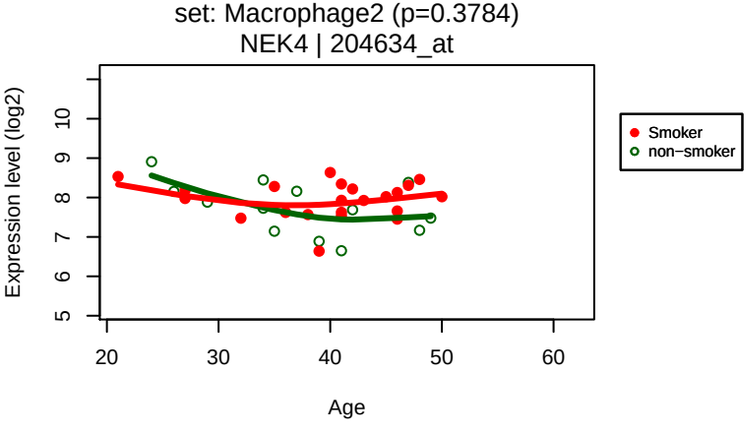

set: SmallAirway2 (p=0.1565)
NEK4 | 204634_at
Expression level (log2)
Age
Smoker
non−smoker

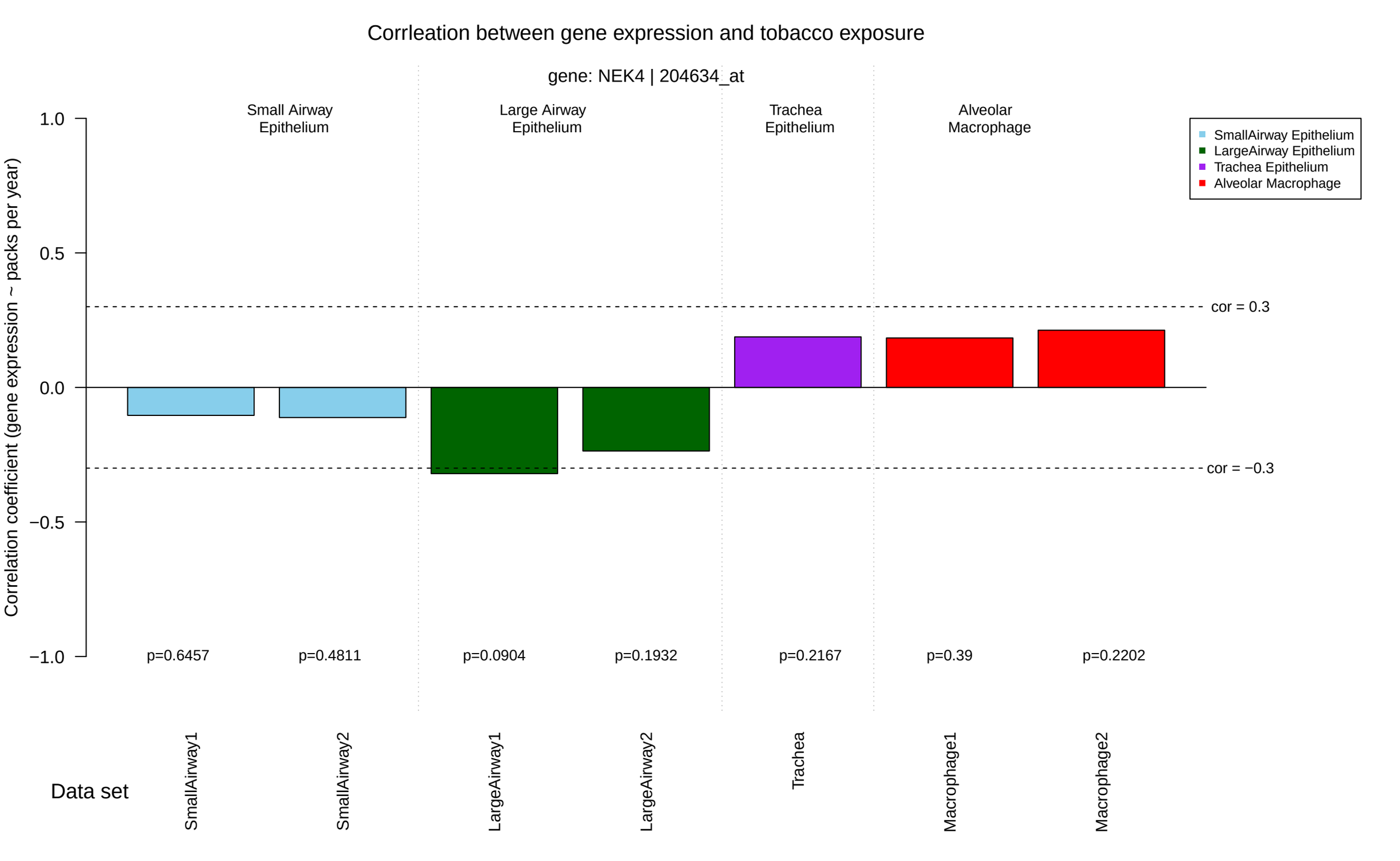

set: LargeAirway1 (p=0.0178)
NEK4 | 204634_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1076)
NEK4 | 204634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5268)
NEK4 | 204634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.942)
NEK4 | 204634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3784)
NEK4 | 204634_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK4 | 204634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6457
p=0.4811
p=0.0904
p=0.1932
p=0.2167
p=0.39
p=0.2202
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2